

Design Considerations

- ❑ What activation function should be used?
- ❑ How many inputs does the network need?
- ❑ How many hidden layers does the network need?
- ❑ How many hidden neurons per hidden layer?
- ❑ How many outputs should the network have?

There is no standard methodology to determinate these values. Even there is some heuristic points, final values are determinate by a **trial and error** procedure.

Pre-processing

- ❑ Transform data to NN inputs
 - ❑ Applying a mathematical or statistical function
 - ❑ Encoding textual data from a database
- ❑ Selection of the most relevant data and outlier removal
- ❑ Minimizing network inputs
 - ❑ Feature extraction
 - ❑ Principal components analysis
 - ❑ Waveform / Image analysis
- ❑ Coding pre-processing data to network inputs